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Page No.:
Date

ECO301 (Assignment - 2)

Q1) Given $u(x_1, x_2) = x_1(x_2+1)$
$= x_1x_2 + x_1$

$P_1 = \$2 \qquad P_2 = \$1$
$m = \$10$

Budget Eq^n $\rightarrow$ $2x_1 + x_2 = 10$

$L = x_1(x_2+1) + \lambda(2x_1 + x_2 - 10)$

$\frac{\partial L}{\partial x_1} = x_2 + 1 + 2\lambda = 0$ — (1)

$\frac{\partial L}{\partial x_2} = x_1 + \lambda = 0$ — (2)

$\frac{\partial L}{\partial \lambda} = 2x_1 + x_2 - 10 = 0$ — (3)

from (1) and (2)

$x_2 + 1 - 2x_1 = 0$
$2x_1 + x_2 = 10$ (from (3))
$2x_2 = 9$
$\boxed{x_2 = 9/2} \approx 4$

$\boxed{x_1 = -\lambda = 11/4} \approx 2$

On doubling the income, budget eqn becomes

$$2x_1 + x_2 = 20 \quad \text{--- (3)}$$

$$L = x_1(x_2+1) - \lambda'(2x_1 + x_2 - 20)$$

$$\frac{\partial L}{\partial x_1} = x_2 + 1 + 2\lambda' = 0$$

$$\frac{\partial L}{\partial x_2} = x_1 + \lambda' = 0$$

$$\frac{\partial L}{\partial \lambda} = 2x_1 + x_2 - 20 = 0$$

On solving (1), (2) and (3) we get

$\lambda = -21/4$ $\boxed{x_1 = 21/4}$ $\boxed{x_2 = 19/2}$

The quantity demanded does not double on doubling the income.

$x_1 = \frac{21}{4} \simeq 5$ $x_2 = 19/2 \simeq 9.9$

Method 2 (Same as week)

Q1) Max utility function $u(x_1, x_2) = x_1(x_2+1)$

price of good 1 = p_1 = \$2

price of good 2 = p_2 = \$1

Income (m) = \$10

Budget eq^n

$$p_1 x_1 + p_2 x_2 = M$$

$$2x_1 + x_2 = 10 \quad -(1)$$

$MU_1 = x_2 + 1$ $\quad MU_2 = x_1$

$$MRS = \frac{-MU_1}{MU_2} = -\frac{(x_2+1)}{x_1}$$

slope of budget equation $= -p_1/p_2$

$= -2$

We can find his demand by equating MRS and budget equation

$$MRS = -p_1/p_2 \Rightarrow -\frac{(x_2+1)}{x_1} = -2$$

So, $2x_1 - x_2 = 1$

and $2x_1 + x_2 = 10$

$x_1 = 11/4$

$x_2 = 11/2 - 1 = 9/2$

Page No.
Date

Demand for good$_1$ = 11/4
" " good 2 = 9/2

(b) If his income doubles
new budget eqn ⇒ $2x_1 + x_2 = 20$

$x_1 = 21/4$ $x_2 = 17/2$

The quantities demanded do not double because of the MRS equation. after doubling the values the MRS is changed and is not equal to the price ratio anymore which is the same as before.

Q2) (i) $U(x_1, x_2) = 2x_1^{0.3} x_2^{0.3} + 1$

$M = \$20$

$p_1 = p_2 = \$1$

Budget Eqn $= x_1 + x_2 = 20$ — (1)

$$MRS = -\frac{\partial U}{\partial x_1} \Big/ \frac{\partial U}{\partial x_2} = -\frac{p_1}{p_2}$$

$$-1 = \frac{\cancel{2(0.3)}\, x_1^{-0.7} x_2^{0.3}}{\cancel{2(0.3)}\, x_2^{-0.7} x_1^{0.3}} = -\frac{x_2^{(0.3+0.7)}}{x_1^{(0.3+0.7)}}$$

$$\frac{x_2}{x_1} = 1 \Rightarrow x_2 = x_1 \quad \text{— (2)}$$

Put (2) in (1)

$2x_1 = 20$

$$\boxed{x_1 = x_2 = 10}$$

Hence, 10 units of each good will be demanded.

(ii) New budget eqn

$\Rightarrow 2x_1 + x_2 = 20$ — (3)

utility function is unaffected

$MRS = -\frac{x_2}{x_1}$ from part (i)

MRS = price ratio $= -\frac{p_1}{p_2}$

$$\frac{-x_2}{x_1} = \frac{-2}{1}$$

$$x_2 = 2x_1 \quad - (4)$$

Put (4) in in 3

$$2x_1 + 2x_1 = 20 \quad \Rightarrow 4x_1 = 20$$

$$\boxed{\begin{aligned} x_1 &= 5 \\ x_2 &= 10 \end{aligned}}$$

Hence, 5 units of good 1 and 10 units of good 2 are demanded.

(iii) His new income $= 20 + 1 \times 5$
$= \$25$

New ~~budget~~ budget Eq^n: $2x_1 + x_2 = 25$ - (5)
(utility function is the same)
and prices are same

So, put (3) in (5)

$$2x_1 + 2x_1 = 25$$
$$x_1 = 25/4 \approx 6$$
$$x_2 = 2x_1 = 25/2 \approx 12$$

$$U(6,12) = (2)(6)^{0.3}(12)^{(0.3)} + 1 = 8.21$$

$$U(10,10) = 2 \times 10^{0.6} + 1 \cancel{= 8.21}$$
$$= 8.962$$

$$U(10,10) > U(6,12)$$

$\therefore$ He won't be better off with tax

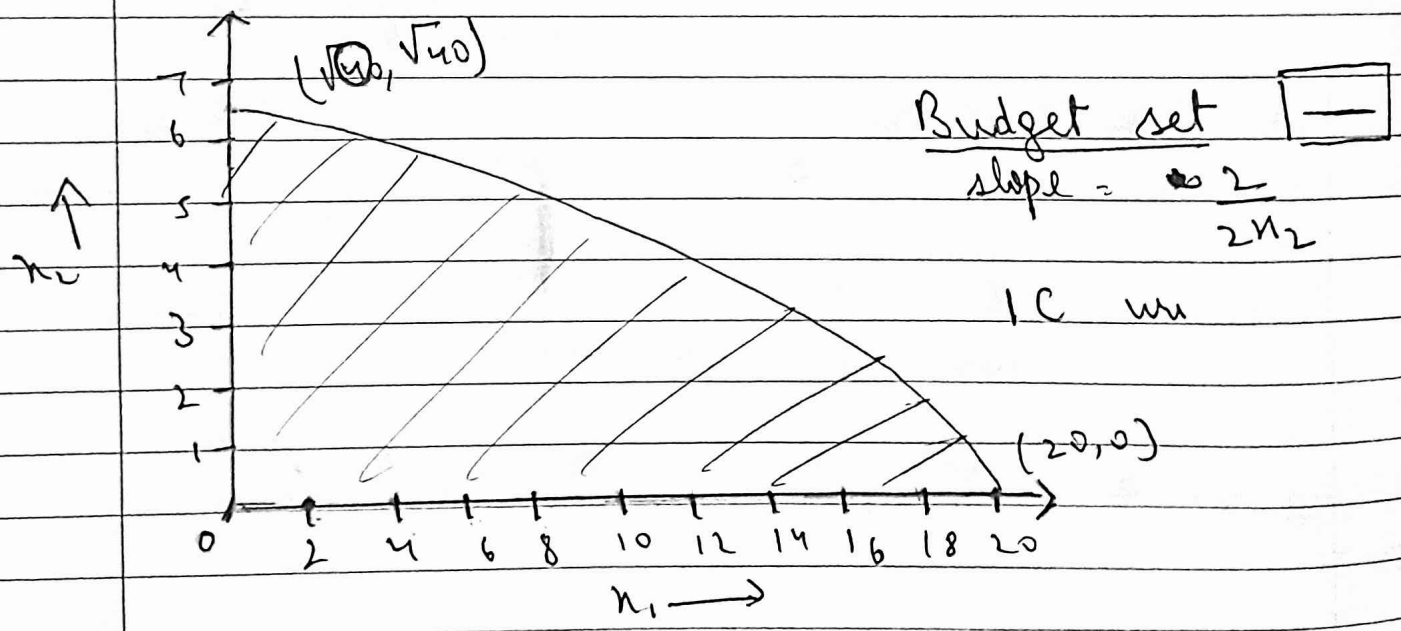
Q3) Let good 2 = cocoa
and good 1 = cheese

$p_2(x_2) = x_2^2$
$p_1 = \$2$
$m = \$40$

$u(x_1, x_2) = x_1 + 2x_2$

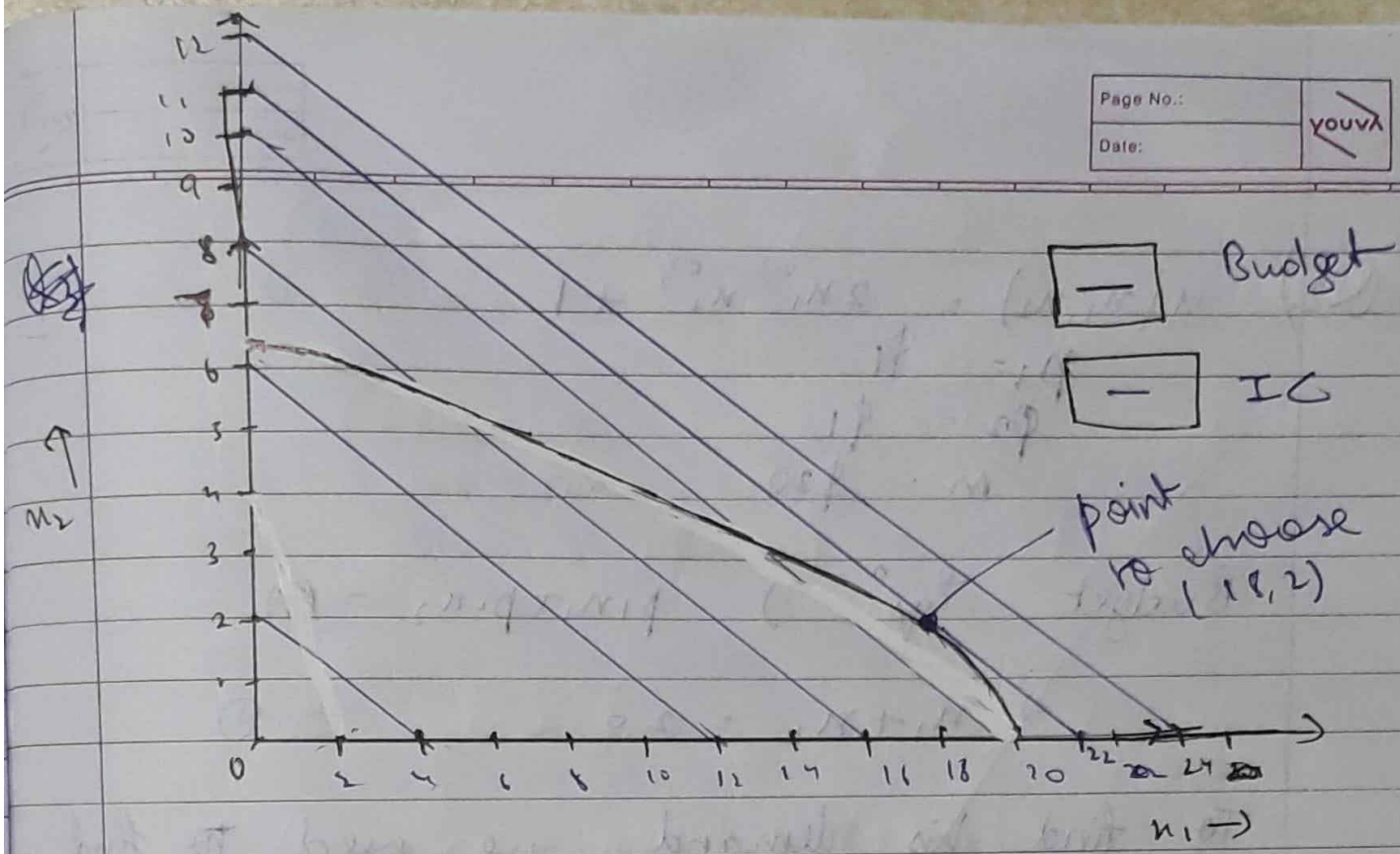
Budget eqn $\Rightarrow$ $2x_1 + x_2^2 = 40$

Some points $\Rightarrow$ $(0, \sqrt{40})$, $(20, 0)$, $(12, 4)$, $(2, 6)$, $(18, 2)$, $(19, 1)$



Adding IC to the same curves.

IC $\Rightarrow$ $x_1 + 2x_2 = K$ [$\because$ K is constant



(c) Budget set $= 2x_1 + x_2^2 \leq 40$

$$L = x_1 + 2x_2 + \lambda(2x_1 + x_2^2 - 40)$$

$$\frac{\partial L}{\partial x_1} = 2\lambda + 1 \quad \text{— (1)}$$

$$\frac{\partial L}{\partial x_2} = 2 + 2\lambda x_2 = 0 \quad \text{— (2)}$$

$$\frac{\partial L}{\partial \lambda} = 2x_1 + x_2^2 - 40 \quad \text{— (3)}$$

Solving (1), (2), (3)

$\lambda = -1/2$, $x_2 = 2$, $x_1 = 18$

Hence, $\boxed{x_1 = 18, \quad x_2 = 2}$

Jasper demands 18 units of cheese and 2 units of cocoa

$U(18, 2) = 18 + 4 = \underline{22}$